Homework 10

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Section 003
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Collaboration Statement

I worked on this homework independently. No outside collaborators or resources were used.

Contents

Problem 1 (*Ross Chapter 3 18.5 (a)*)

Let f and g be continuous functions in $[a, b]$ such that $f(a) \geq g(a)$ and $f(b) \leq g(b)$. Prove $f(x_0) = g(x_0)$ for at least one x_0 in $[a, b]$.

Discussion.
We want a point where $f(x_0) = g(x_0)$. This is equivalent to $f(x_0) - g(x_0) = 0$, so it is natural to consider the difference $h(x) = f(x) - g(x)$. Since f and g are continuous, h is continuous on $[a, b]$. The inequalities $f(a) \geq g(a)$ and $f(b) \leq g(b)$ then say $h(a) \geq 0$ and $h(b) \leq 0$, so 0 lies between $h(a)$ and $h(b)$. By the Intermediate Value Theorem applied to h, there is some $x_0 \in [a, b]$ with $h(x_0) = 0$, i.e. $f(x_0) = g(x_0)$.

Proof.
Define $h : [a, b] \to \mathbb{R}$ by

$$h(x) = f(x) - g(x).$$

Since f and g are continuous on $[a, b]$ and the difference of two continuous functions is continuous, it follows that h is continuous on $[a, b]$.

From the hypotheses we have

$$f(a) \geq g(a) \quad \Longrightarrow \quad h(a) = f(a) - g(a) \geq 0,$$

and

$$f(b) \leq g(b) \quad \Longrightarrow \quad h(b) = f(b) - g(b) \leq 0.$$

Thus 0 lies between $h(a)$ and $h(b)$; equivalently, $h(b) \leq 0 \leq h(a)$.

By the Intermediate Value Theorem, since h is continuous on $[a, b]$ and 0 lies between $h(a)$ and $h(b)$, there exists some $x_0 \in [a, b]$ such that

$$h(x_0) = 0.$$

By the definition of h this means

$$f(x_0) - g(x_0) = 0,$$

so $f(x_0) = g(x_0)$.

Therefore there exists at least one $x_0 \in [a, b]$ with $f(x_0) = g(x_0)$, as required. $\square$

Problem 2 (*Ross Chapter 3 18.10*)

Suppose f is continuous on $[0, 2]$ and $f(0) = f(2)$. Prove there exist x, y in $[0, 2]$ such that $|y - x| = 1$ and $f(x) = f(y)$. *Hint*: Consider $g(x) = f(x + 1) - f(x)$ on $[0, 1]$.

Discussion.
We want two points x, y with $|y - x| = 1$ and $f(x) = f(y)$. Writing $y = x + 1$, this is equivalent to finding $x \in [0, 1]$ such that

$$f(x + 1) - f(x) = 0.$$

So it is natural to define

$$g(x) = f(x + 1) - f(x), \text{ for } x \in [0, 1].$$

Because f is continuous on $[0, 2]$, both $f(x + 1)$ and $f(x)$ are continuous on $[0, 1]$, so their difference g is continuous on $[0, 1]$. The condition $f(0) = f(2)$ then forces

$$g(0) = f(1) - f(0), \qquad g(1) = f(2) - f(1) = f(0) - f(1) = -g(0),$$

so either $g(0) = 0$ or $g(0)$ and $g(1)$ have opposite signs. In either case, 0 lies between $g(0)$ and $g(1)$, so by the Intermediate Value Theorem there is some $x \in [0, 1]$ with $g(x) = 0$, i.e. $f(x + 1) = f(x)$ and $|(x + 1) - x| = 1$.

TL;DR: We basically have to translate the statement we are trying to prove to make it in terms of just x (which we do by defining g) to then make the question about applying the IVT to g to prove that it has a zero, which then maps back to the original statement we are trying to prove about f.

Proof.
Let g be a function defined on $[0, 1]$ as

$$g(x) = f(x + 1) - f(x).$$

Since f is continuous on $[0, 2]$ (and $[0, 1] \subset [0, 2]$), then g must be continuous on $[0, 1]$.

We compute the endpoint values of g:

$$g(0) = f(1) - f(0),$$

and, using $f(0) = f(2)$,

$$g(1) = f(2) - f(1) = f(0) - f(1) = -g(0).$$

If $g(0) = 0$, then taking $x = 0$ and $y = 1$ gives

$$|y - x| = |1 - 0| = 1, \qquad f(x) = f(0) = f(1) = f(y),$$

so the desired property holds.

Otherwise $g(0) \neq 0$, so $g(1) = -g(0)$ has the opposite sign. Thus

$$g(0) \cdot g(1) < 0,$$

which means 0 lies strictly between $g(0)$ and $g(1)$. Since g is continuous on $[0, 1]$, the Intermediate Value Theorem guarantees the existence of some $x \in (0, 1)$ such that

$$g(x) = 0.$$

For this x, set $y = x + 1$. Then $x, y \in [0, 2]$, we have

$$|y - x| = |x + 1 - x| = 1,$$

and

$$f(y) = f(x + 1) = f(x),$$

because $g(x) = f(x + 1) - f(x) = 0$.

Therefore there exist $x, y \in [0, 2]$ with $|y - x| = 1$ and $f(x) = f(y)$, as required. $\square$

Problem 3 (*Ross Chapter 3 19.6 (b)*)

(From Part (a):) Let $f(x) = \sqrt{x}$ *for* $x \geq 0$. *Show* f' *is unbounded on* $(0, 1]$ *but* f *is nevertheless uniformly continuous on* $(0, 1]$ *Compare with Theorem 19.6.*

(b) Show that f is uniformly continuous on $[1, \infty)$.

Discussion.

We want to prove that $f(x) = \sqrt{x}$ is *uniformly* continuous on $[1, \infty)$. By Definition 19.1, this means:

$$\text{for each } \varepsilon > 0 \text{ there exists } \delta > 0 \text{ such that } x, y \in [1, \infty),\ |x - y| < \delta \Longrightarrow |\sqrt{x} - \sqrt{y}| < \varepsilon.$$

So the game is: fix an arbitrary $\varepsilon > 0$ and then cook up a δ that depends *only* on ε, not on a particular choice of x or y.

A standard trick for square roots is to use the algebraic identity

$$|\sqrt{x} - \sqrt{y}| = \left| \frac{x - y}{\sqrt{x} + \sqrt{y}} \right|.$$

On our domain $[1, \infty)$ we always have $\sqrt{x} \geq 1$ and $\sqrt{y} \geq 1$, so

$$\sqrt{x} + \sqrt{y} \geq 2.$$

That means we can bound the difference by

$$|\sqrt{x} - \sqrt{y}| \leq \frac{|x - y|}{2}.$$

This is the key estimate: on $[1, \infty)$, the function $\sqrt{x}$ cannot change faster than "slope $\frac{1}{2}$" in the sense that

$$|\sqrt{x} - \sqrt{y}| \leq \frac{1}{2}\,|x - y|.$$

Once we have this, the ε–δ choice is easy: if we make $|x - y|$ smaller than 2ε, then $\frac{1}{2}|x - y|$ will automatically be smaller than ε. So in the proof we will take

$$\delta = 2\varepsilon$$

and then check that this choice satisfies the definition of uniform continuity on $[1, \infty)$.

Proof.

Let $\varepsilon > 0$, and $\delta = 2\varepsilon$. From Definition 19.1 let $x, y \in S$ with $|x - y| < \delta$. Then,

$$|\sqrt{x} - \sqrt{y}| = \left| \frac{x - y}{\sqrt{x} + \sqrt{y}} \right|.$$

Because $x, y \in [1, \infty)$, we have $\sqrt{x} \geq 1$ and $\sqrt{y} \geq 1$, so

$$\sqrt{x} + \sqrt{y} \geq 2.$$

Therefore

$$|\sqrt{x} - \sqrt{y}| = \frac{|x-y|}{\sqrt{x} + \sqrt{y}} \leq \frac{|x-y|}{2}.$$

Using $|x - y| < \delta = 2\varepsilon$, we obtain

$$|\sqrt{x} - \sqrt{y}| \leq \frac{|x-y|}{2} < \frac{2\varepsilon}{2} = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, by Definition 19.1, $f(x) = \sqrt{x}$ is uniformly continuous on $[1, \infty)$. $\square$

Problem 4 (*Ross Chapter 3 19.7*)

(a) Let f be a continuous function on $[0, \infty)$. Prove that if f is uniformly continuous on $[k, infty)$ for some k, then f is uniformly continuous on $[0, \infty)$.

(b) Use (a) and Exercise 19.6(b) to show that $f(x) = \sqrt{x}$ is uniformly continuous on $[0, \infty)$.

Discussion.

Proof.

□

Problem 5 (*Uniform Continuity of* f)

Suppose $f : \mathbb{R} \to \mathbb{R}$. Prove that if there exists a real number $M \geq 0$ such that $f(x) - f(y)| \leq M|x - y|$ for all $x, y \in \mathbb{R}$, then f is uniformly continuous on $\mathbb{R}$.

Discussion.

Proof.

$\square$

Problem 6 (*Differntiability of* f)

Suppose that $0 \in (a, b)$ and $f(x) = xg(x)$ for some function $g : (a, b) \to \mathbb{R}$ which is continuous at $x = 0$. Prove that f is differentiable at $x = 0$, and find $f'(0)$ in terms of g.

Hint: Note that you cannot use the product rule, since $g(x)$ is not guaranteed to be differentiable at $x = 0$

Discussion.

Proof.

$\square$

Appendix: Theorems

Theorem 17.2

Let f be a real-valued function whose domain is a subset of $\mathbb{R}$. Then f is continuous at $x_0 \in \mathrm{dom}(f)$ if and only if

$$\text{for each } \varepsilon > 0 \text{ there exists } \delta > 0 \text{ such that } x \in \mathrm{dom}(f) \text{ and}$$
$$|x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon.$$

Theorem 18.1 — Extreme Value Theorem

Let f be a continuous real-valued function on a closed interval $[a, b]$. Then f is a bounded function. Moreover, f assumes its maximum and minimum values on $[a, b]$; that is, there exist x_0, y_0 in $[a, b]$ such that

$$f(x_0) \leq f(x) \leq f(y_0) \quad \text{for all } x \in [a, b].$$

Theorem 18.2 — Intermediate Value Theorem

If f is a continuous real-valued function on an interval I, then f has the intermediate value property on I: Whenever $a, b \in I$, $a < b$, and y lies between $f(a)$ and $f(b)$ (i.e., $f(a) < y < f(b)$ or $f(b) < y < f(a)$), there exists at least one x in (a, b) such that

$$f(x) = y.$$

Corollary 18.3

If f is a continuous real-valued function on an interval I, then the set

$$f(I) = \{f(x) : x \in I\}$$

is also an interval or a single point.

Definition 19.1 — Uniform Continuity

Let f be a real-valued function defined on a set $S \subseteq \mathbb{R}$. Then f is *uniformly continuous* on S if

$$\text{for each } \varepsilon > 0 \text{ there exists } \delta > 0 \text{ such that } x, y \in S \text{ and } |x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon.$$

We will say f is *uniformly continuous* if f is uniformly continuous on $\mathrm{dom}(f)$.

Theorem 19.2

If f is continuous on a closed interval $[a, b]$, then f is uniformly continuous on $[a, b]$.

Discussion 19.3

Example 5 illustrates the power of Theorem , but it still may not be clear why uniform continuity is worth studying. One of the important applications of uniform continuity concerns the integrability of continuous functions on closed intervals.
To see the relevance of uniform continuity, consider a continuous nonnegative real-valued function f on $[0, 1]$. For $n \in \mathbb{N}$ and $i = 0, 1, 2, \ldots, n-1$, let

$$M_{i,n} = \sup\{f(x) : x \in [\tfrac{i}{n}, \tfrac{i+1}{n}]\}, \qquad m_{i,n} = \inf\{f(x) : x \in [\tfrac{i}{n}, \tfrac{i+1}{n}]\}.$$

Then the upper sum and lower sum are given by

$$U_n = \frac{1}{n}\sum_{i=0}^{n-1} M_{i,n}, \qquad L_n = \frac{1}{n}\sum_{i=0}^{n-1} m_{i,n}.$$

The function f would be Riemann integrable provided $U_n - L_n \to 0$, i.e.,

$$\lim_{n\to\infty}(U_n - L_n) = 0.$$

Since f is uniformly continuous on $[0, 1]$ by Theorem , there exists $\delta > 0$ such that

$$x, y \in [0, 1],\ |x - y| < \delta \quad \Rightarrow \quad |f(x) - f(y)| < \varepsilon.$$

Choosing N such that $\frac{1}{N} < \delta$ and $n > N$ gives

$$0 \le U_n - L_n = \frac{1}{n}\sum_{i=0}^{n-1}(M_{i,n} - m_{i,n}) < \varepsilon.$$

This proves the claim. The next two theorems show uniformly continuous functions have nice properties.

Definition 28.1 — Differentiability

Let f be a real-valued function defined on an open interval that contains a point a. We say that f is *differentiable at* a, or that f *has a derivative at* a, if the limit

$$\lim_{x\to a}\frac{f(x) - f(a)}{x - a}$$

exists (as a finite real number).

Theorem 28.2

If f is differentiable at a point a, then f is continuous at a.

Theorem 28.3 — Algebra Rules

Let f and g be functions that are differentiable at a point a, and let c be a real constant. Then each of the following functions is differentiable at a (whenever it is defined), and their derivatives are given by:

(a) Constant multiple:

$$(cf)'(a) = c \cdot f'(a).$$

(b) Sum:

$$(f+g)'(a) = f'(a) + g'(a).$$

(c) Product rule:

$$(fg)'(a) = f(a)g'(a) + f'(a)g(a).$$

(d) Quotient rule: If $g(a) \neq 0$, then f/g is differentiable at a and

$$\left(\frac{f}{g}\right)'(a) = \frac{g(a)f'(a) - f(a)g'(a)}{[g(a)]^2}.$$

Theorem 28.4 — Chain Rule

Assume f is differentiable at a and g is differentiable at $f(a)$. Then the composite function $g \circ f$ is differentiable at a, and

$$(g \circ f)'(a) = g'\big(f(a)\big)\, f'(a).$$